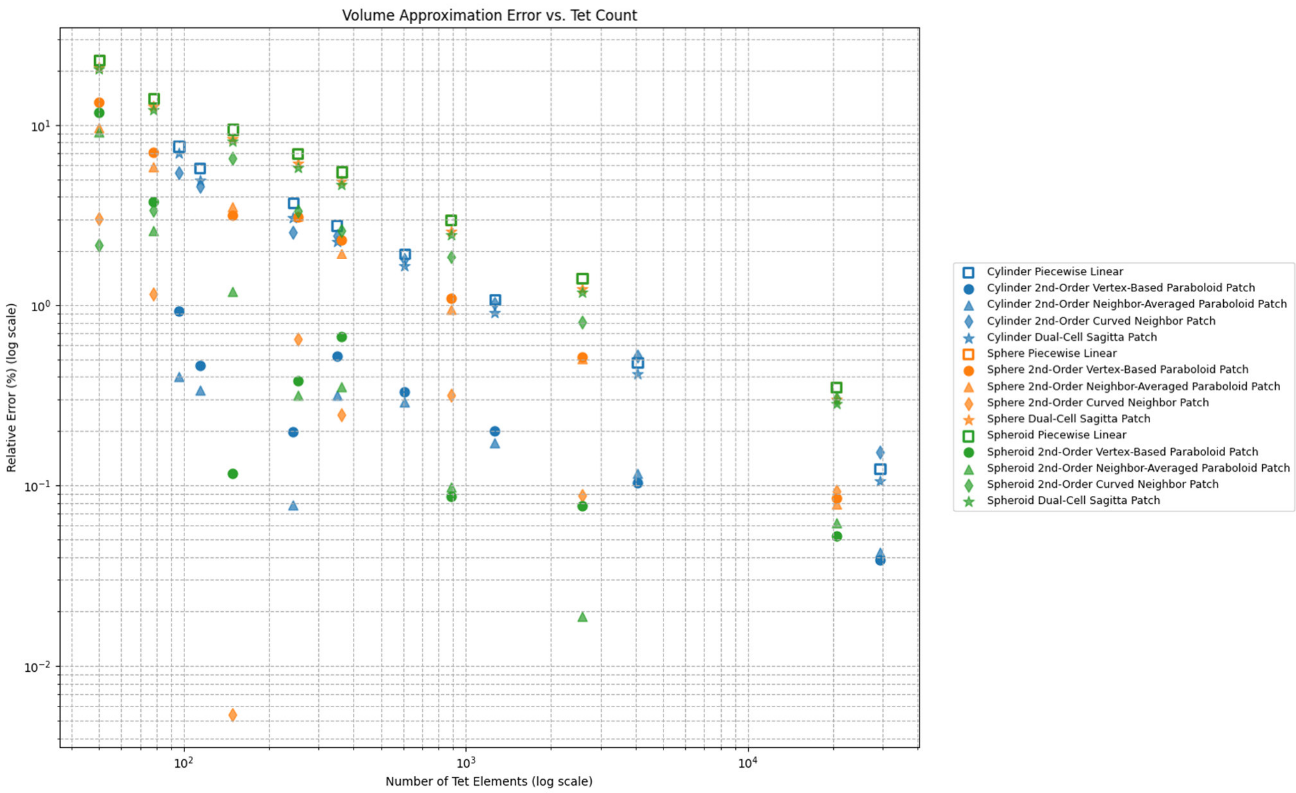

Equations for Dual-Cell Sagitta Patch

1. Dual Cell Construction

- **Dual Cell** (Voronoi cell) for vertex i:
  Polygon formed by the circumcenters of all triangles sharing vertex i.

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2. Dual Area

- **Polygon area in 3D:**
  If the circumcenters are $\{\mathbf{c}_k\}$ ordered counterclockwise,
  and $\mathbf{n}$ is the average normal:

$$A_i^* = \text{polygon\_area\_3d}(\{\mathbf{c}_k\}, \mathbf{n})$$

  (in code, this is the `polygon_area_3d` function using the 2D projection and shoelace formula).

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3. Curvature at Vertex

- **Mean curvature at vertex:**

$$H_i = (\text{estimated at vertex } i \text{ using DDG/mesh Laplacian})$$

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4. Sagitta Calculation

- **Sagitta** (height of paraboloid above base polygon, with average radius r and curvature radius R):
  - **Curvature radius:** $R = \frac{1}{H_i}$
  - **Average dual cell radius:**

$$r = \text{mean}\left(\|\mathbf{c}_k - \mathbf{v}_i\|\right)$$

  - **Sagitta formula:**

$$s = R - \sqrt{R^2 - r^2} \quad (\text{if } R^2 > r^2 \text{ and } R > 0, \text{ else } 0)$$

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5. Dual-Cell Sagitta Volume

- **Volume contribution at vertex i:**

$$V_{\text{dual-sagitta}}^{(i)} = \frac{1}{3} A_i^* \cdot s$$

where s is the sagitta, A_i^* is the dual cell area.

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6. Total Volume Correction

- **Sum over all vertices:**

$$V_{\text{total,dual-sagitta}} = V_{\text{flat}} + \sum_i V_{\text{dual-sagitta}}^{(i)}$$

where V_{flat} is the total piecewise-linear mesh volume.

Full Summary Table

Step	Equation	In code
Dual area	$A_i^* = \text{polygon\_area\_3d}(\{\mathbf{c}_k\}, \mathbf{n})$	`polygon_area_3d`
Curvature radius	$R = \frac{1}{H_i}$	`R = 1.0 / H`
Dual cell mean radius	$r = \text{mean}(\|\mathbf{c}_k - \mathbf{v}_i\|)$	`r = np.mean(radii)`
Sagitta	$s = R - \sqrt{R^2 - r^2}$	`sagitta = R - np.sqrt(val)`
Patch volume	$V_{\text{dual-sagitta}}^{(i)} = \frac{1}{3} A_i^* s$	`V_sag = (1.0/3.0)*dual_area*sagitta`
Total volume	$V_{\text{total,dual-sagitta}} = V_{\text{flat}} + \sum_i V_{\text{dual-sagitta}}^{(i)}$	`total_V_dual_sagitta`

- Function: `dual_cell_sagitta_patch(...)`
- Area: `dual_area = polygon_area_3d(...)`
- Sagitta: `sagitta = R - np.sqrt(val)` where `val = R**2 - r**2`
- Volume: `V_sag = (1.0 / 3.0) * dual_area * sagitta`
- Sum for all vertices, add to flat volume for total.